

Hybrid Agentic Workflows using LangGraph

By: Athreya Kidambi

1. Introduction

This assignment focuses on designing hybrid agentic workflows using LangGraph across three real-world domains—Telecom, Healthcare, and Finance. The goal is to simulate intelligent systems that can understand user inputs, enrich them using backend tools, dynamically route requests, and generate structured, actionable outputs.

Each workflow follows a consistent architecture built on LangGraph, combining LLM-based reasoning, tool integration, and rule-based routing. The implementation demonstrates how multi-agent systems can be orchestrated to solve domain-specific problems such as network incident resolution, clinical triage, and loan approval automation.

2. Overall Architecture

All three workflows adhere to a four-stage design:

Stage 1 – Understanding:

The system interprets user input using structured LLM outputs. Depending on the domain, it extracts attributes such as issue category, urgency, symptoms, or applicant details.

Stage 2 – Backend Enrichment:

Mock tools are used to simulate real-world backend systems. These tools provide additional context such as network status, medical guidelines, or financial verification data.

Stage 3 – Intelligent Routing:

Using conditional edges in LangGraph, the system dynamically routes the request to specialized agents based on the interpreted input and enriched data.

Stage 4 – Resolution Generation:

Each specialized agent produces a structured output tailored to the domain, including end-user responses and operational insights.

This architecture reflects a hybrid approach, combining LLM reasoning with deterministic control logic to ensure both flexibility and reliability.

3. Telecom Domain – Network Incident Resolution

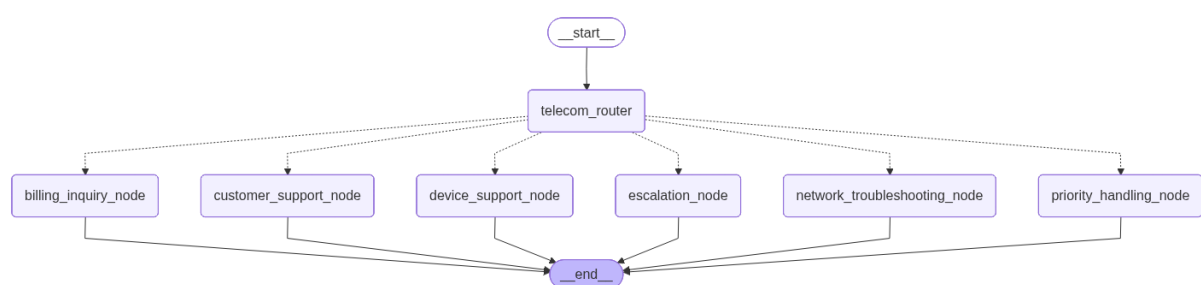
The telecom workflow handles customer complaints related to network services. It begins with a router node that classifies the complaint based on category, severity, and impact scope. An additional rule detects repeated complaints and prioritizes them for special handling.

Based on this classification, the system routes the request to specialized nodes such as customer support, network troubleshooting, billing, device support, escalation, or priority handling. High-severity or large-scale issues are immediately escalated, while repeated complaints are routed to a dedicated priority handling node.

Backend tools simulate telecom systems, including tower health checks, outage monitoring, billing records, and CRM tickets. These enrich the agent’s reasoning and enable more realistic responses.

The output includes three components: a customer-facing response, internal technician notes, and an estimated resolution time. This ensures both external communication and internal operational clarity.

Example scenarios demonstrate the workflow’s effectiveness. A general connectivity issue is escalated due to high severity, a device compatibility issue is resolved through troubleshooting steps, a city-wide outage triggers emergency escalation, and repeated complaints result in prioritized resolution with technician dispatch.



4. Healthcare Domain – Clinical Triage System

The healthcare workflow simulates a clinical triage system that processes patient queries and provides appropriate care recommendations.

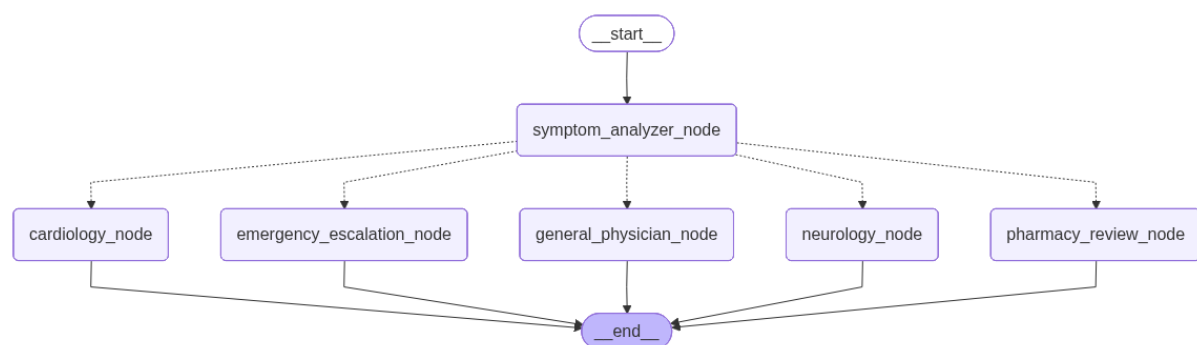
The workflow begins with a symptom analyzer node, which extracts symptoms, determines urgency, detects emergency conditions, and recommends a specialist area. This structured understanding ensures accurate downstream routing.

The system then uses medical tools to retrieve guidelines, check drug interactions, fetch specialist information, and access patient history where available. These tools help simulate real-world healthcare decision support.

Routing is performed based on urgency and specialization. Emergency cases are handled immediately, while non-critical cases are directed to specialists such as cardiologists, neurologists, pharmacists, or general physicians.

Each specialized agent generates a triage summary, specialist recommendation, and follow-up instructions, along with diagnostic suggestions when necessary. For critical cases, the system explicitly advises contacting emergency services.

The workflow effectively handles diverse scenarios. Emergency symptoms such as chest pain trigger immediate escalation, general symptoms are handled with physician recommendations, drug interactions are flagged through pharmacy review, and specialist-specific cases incorporate patient history for more contextual guidance.



5. Finance Domain – Loan Risk Assessment

The finance workflow implements an end-to-end system for loan application evaluation.

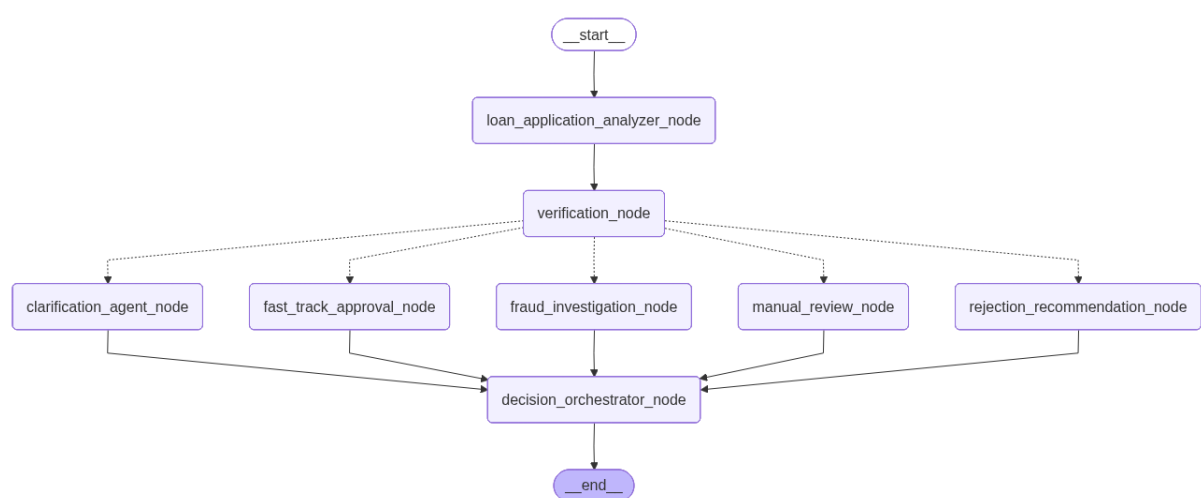
The process begins with structured parsing of the loan application using a Pydantic schema. Applicant details such as credit score, employment type, and document availability are extracted and analyzed. The system evaluates income stability, detects missing documents, and provides an initial financial assessment.

In the next stage, verification tools simulate backend checks such as credit score validation, employment verification, fraud risk scoring, KYC verification, and debt-to-income (DTI) calculation. These checks provide critical information for decision-making.

The routing logic then determines the appropriate action. High fraud risk leads to escalation, missing documents result in clarification requests, strong profiles receive fast-track approval, and weaker profiles are routed for manual review or rejection.

A final orchestrator node consolidates the decision into a structured output that includes the loan decision, rationale, risk summary, recommended amount, interest slab, and applicant communication.

Example scenarios illustrate the workflow clearly. A high-credit applicant is fast-tracked for approval, an applicant with missing documents is asked for clarification, and moderate-risk cases are handled through manual review. The system effectively balances automation and risk control.



6. Implementation Practices

The design emphasizes modularity, with each agent node performing a clearly defined function. Structured outputs using Pydantic models ensure consistency and reliability across workflows.

7. Key Observations

The implementation highlights the importance of combining LLM reasoning with structured workflows. While the LLM provides flexibility in interpreting inputs, the rule-based routing ensures predictable and safe system behaviour.

Each domain demonstrates a different aspect of agentic workflows: telecom focuses on operational escalation, healthcare emphasizes safety-critical triage, and finance showcases risk-based decision-making. Together, they illustrate how LangGraph enables scalable multi-agent orchestration across diverse business scenarios.

8. Conclusion

This assignment successfully demonstrates the design and implementation of hybrid agentic workflows using LangGraph across three domains. The workflows show how intelligent systems can interpret inputs, interact with backend tools, dynamically route tasks, and generate meaningful outputs.

The Telecom, Healthcare, and Finance solutions collectively reflect real-world applications of AI-driven orchestration, highlighting the potential of modular, multi-agent architectures in enterprise systems.